

Classification of postures of people with hand movement working in office environment by LiDAR sensor network

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ABSTRACT An edge computing system is proposed for classifying postures with hand movement of office workers using a light detection-and-ranging (LiDAR) sensor network. The system acquires human poses with hand movement through the LiDAR sensor network, applies preprocessing including trimming, feature estimation, clustering, and normalization, and classifies the postures by deep learning.

The system is evaluated on posture data from multiple LiDAR sensors and multiple subjects under varied conditions of data type, LiDAR placement, features, and clustering methods. Classification accuracy and confusion matrices are reported for the trained models.

INDEX TERMS posture classification, light detection and ranging, point cloud, deep learning.

I. INTRODUCTION

THE office-efficiency monitoring has been attracting people [1]–[6]. A recent report lists productivity monitoring, project tracking, and employee monitoring as its main functions [1].

Office-efficiency monitoring requires recording time-varying worker activities beyond gross postures or presence. Hand-movement classification is necessary because desk work is performed through hand-involved movements that gross postures cannot distinguish. Fine-grained hand-movement classification supports task-switch identification and actual work-duration estimation.

Prior works focused on identifying human body orientation and general human postures. Katayama et al. reported that point clouds can be utilized to classify human posture and presented a system for posture recognition and tracking [7]. The system classified posture from LiDAR point clouds using machine learning but was evaluated with only one participant. Glandon et al. reported that point clouds can be utilized for human identification and presented a system for posture estimation using data captured from walking people [8]. The system classified posture from joint features extracted via silhouettes of walking subjects. The prior works lack hand-movement classification and models that extract fine point-

cloud details for multiple people.

An edge computing system is proposed that classifies the postures of people with hand movement in an office by using a LiDAR sensor network. The system captures, stores, and classifies human postures with hand movement through the LiDAR sensor network. The system is evaluated on posture data from multiple LiDAR sensors under varied data types, LiDAR placements, features, and clustering methods. Preprocessing including trimming, feature estimation, clustering, and normalization is applied before deep-learning classification. Classification accuracy is measured for models trained under each condition. The evaluation metric is the ratio of correctly classified samples; confusion matrices are presented for selected models.

II. RELATED WORKS

A. MODEL ARCHITECTURE

Classification associates each data sample with the category of highest estimated probability. Deep learning models for point cloud classification have attracted interest [9]. Early approaches relied on voxelization [10]–[12]. Qi et al. presented PointNet, which applies shared multilayer perceptrons (MLPs) to each point and aggregates features with max pooling under permutation invariance [9]. Subsequent models

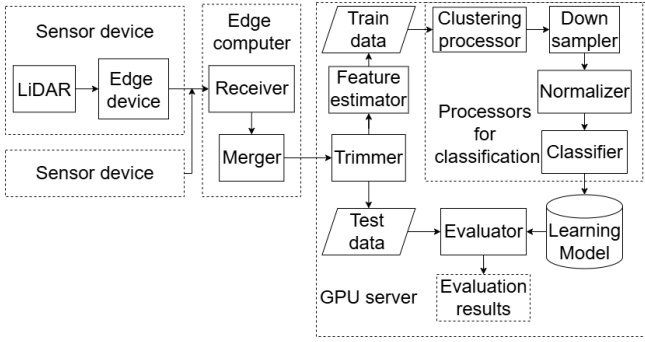


FIGURE 1. System model.

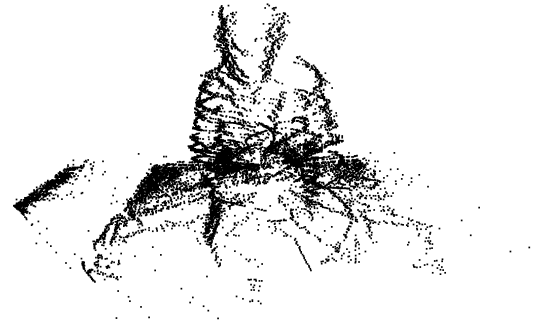


FIGURE 2. Trimmed data.

such as PointCNN [13], ShellNet [14], A-CNN [15], SO-Net [16], and Kd-network [17] learn hierarchical local structure from point coordinates. The existing research comprises direct classification architectures and spatial modeling using local relationships. The present study targets fine-grained hand-movement classification in a multi-person office environment, unlike macro-posture or static object classification.

B. IMPROVEMENT FOR OBJECT DETECTION

Point features encode inter-point relationships and local shape and are used with point coordinates as machine-learning inputs. Eigenvalue-based features from neighborhood covariance matrices include entropy and omnivariance [18]–[20]. Convex-hull descriptors [21] and higher-dimensional descriptors such as signature of histograms of orientations (SHOT) [22] and position and orientation distribution (POD) [23] capture object shape for recognition and retrieval. The present study applies point feature preprocessing to hand-movement classification in a multi-person office environment.

III. PROPOSED SYSTEM

A. SYSTEM MODEL

Fig. 1 shows the system model of the proposed system, which comprises sensor devices, an edge computer, and a GPU server. Each sensor device has a LiDAR unit and an edge device; the edge computer has a receiver and a merger; the GPU server has a trimmer, a feature estimator, and classification processors with clustering, downsampling, normalizing, and classifying parts.

3D data are acquired by each LiDAR unit, stored on the edge device, received and merged by the edge computer, and processed on the GPU server through trimming, feature extraction, clustering, downsampling, normalization, and classification.

B. METHODOLOGY

In the proposed system, 3D data are acquired using the LiDAR network of Akiyama *et al.* and Li *et al.* [24], [25]. Multiple LiDAR units are calibrated for merging by rotation and translation based on predetermined angles and distances. Each frame of the aggregated point cloud is trimmed manu-

ally to the workspace region in a fixed-seating office. Fig. 2 shows a trimmed point cloud. Four feature types are extracted from the trimmed point clouds: normals, dimensionality features, proportion of variance, and fast point feature histograms (FPFH). For neighborhood-based features, the k-nearest neighbor algorithm (KNN) uses search radius 100 and a maximum of 30 neighbors. *Normals*. Eigenvectors of the smallest eigenvalue in each covariance matrix define the normal. Fig. 3(a) shows the data with the normals. *Dimensionality features* [26], [27]. Eigenvalues are defined as λ_i , $i = [1, 3]$ and the dimensionality features are calculated as $a_{1D} = \frac{\sqrt{\lambda_1 - \sqrt{\lambda_2}}}{\sqrt{\lambda_1}}$, $a_{2D} = \frac{\sqrt{\lambda_2 - \sqrt{\lambda_3}}}{\sqrt{\lambda_1}}$, and $a_{3D} = \frac{\sqrt{\lambda_3}}{\sqrt{\lambda_1}}$. Fig. 3(b) shows the data with the dimensionality features. *Proportion of variance* [28]. The proportion of variance is calculated as $v_i = \frac{\lambda_i}{\lambda_1 + \lambda_2 + \lambda_3}$ ($i = [1, 3]$). Fig. 3(c) shows the data with the proportion of variance. *FPFH* [29]. FPFH is computed from Darboux-frame variables between adjacent points and normals [30], [31], and reduced from 33 to three dimensions by principal component analysis (PCA) with a randomized singular value decomposition (SVD) solver. Fig. 3(d) shows the data with the FPFH. Clustering is performed using DBSCAN with $\varepsilon = 100$, minimum 10 points, and three-dimensional (3D) coordinates, or k-means with $k = 3$; outliers and selected clusters are removed [32], [33]. The clustered point clouds are downsampled and normalized for training data with unit distance from the center of gravity to the farthest point. Classification is performed on the normalized point clouds by PointNet++, which twice selects points, groups neighborhoods, and processes each group with PointNet and a fully connected layer [34].

IV. EXPERIMENT SETUP

A. EXPERIMENTAL SYSTEM

An experimental system was developed based on the proposed system in Fig. 1. The system comprised two Livox Mid-70 LiDAR units, two Jetson nano edge devices, Jetson Xavier NX and Orin nano edge computers, and a keyboard [35]–[38]. Table 1 lists the specifications of the experiment.

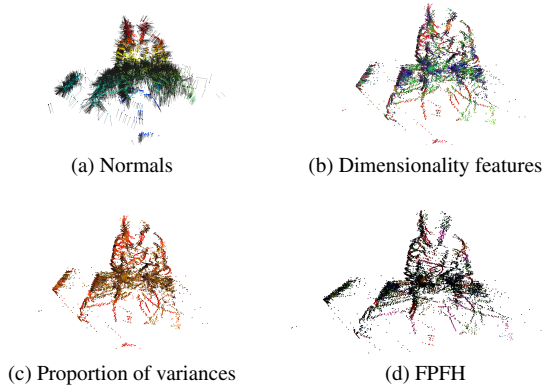


FIGURE 3. Point clouds with each of four types of feature.

B. LIDAR DEPLOYMENT

Fig. 5 shows the deployment of the devices for the experiment. Two LiDAR sensors faced each other on a table, 1.0 m from the subject laterally and tilted 10° downward, with a PC in front of the subject. Fig. 4 shows the experiment site.

C. DATA ACQUISITION

1) Training data acquisition

Training data were acquired for four posture classes—mouse operation, pad operation, sitting still, and typing—while subjects used a PC.¹ Mouse operation, pad operation, and typing were captured for 30 s each with 20 repetitions; sitting still was captured for 10 s with 10 repetitions. Fig. 6 shows example point clouds for each class.

The three subjects in this study are denoted as subjects A, B, and C. Training data with subjects A, B, and C is called training data A, B, and C.

¹The experiment was approved by the Ethics Committee of Shibaura Institute of Technology.

TABLE 1. Specifications of experiment.

Parameter	Value
Place	14Q32, 14F, Shibaura Institute of Technology, Toyosu, Koto-ku, Tokyo
Date of training data and test data A, B, and C acquisition	9 December 2022
Date of test data A to AB acquisition	12 December 2024
LiDAR units	Livox Mid-70
No. of LiDAR units	2
Edge devices	NVIDIA Jetson nano
No. of edge devices	2
Edge computer for training data and test data A, B, and C acquisition	NVIDIA Jetson Xavier NX
Edge computer for test data A to AB acquisition	NVIDIA Jetson Orin nano
Frame rate of sensor	10 fps
No. of contributors for training data	3
No. of contributors for test data	28



FIGURE 4. Experiment site.

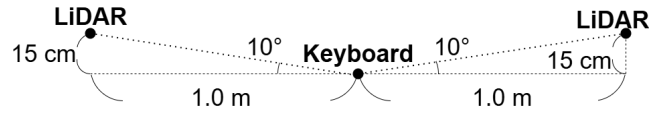


FIGURE 5. Deployment of LiDAR sensors.

2) Test data acquisition

Test data were acquired for the four posture classes under the conditions summarized in Table 2; each capture was performed once per subject.

The twenty-eight subjects in this study are denoted as subjects A to AB. Test data with subjects A to AB is called test data A to AB.

D. PROCESSING OF PROPOSED SYSTEM

Processing followed the methodology in Section III-B. Point clouds were trimmed from manually determined person-region vertices.

Normals, dimensionality features, proportion of variance, and FPFH were computed with Open3D estimators.

DBSCAN and k-means clustering removed outliers and selected clusters.

Each point cloud was downsampled to 4,096 points [39].

Training data were normalized per dataset.

PointCutMix augmentation [40] replaced the top $\lambda\%$ of nearest-neighbor-matched points in cloud A with points from clouds B and C, where λ was drawn from a beta distribution.

Augmented data were split 4:1 for training and validation. PointNet++ grouping used multi-scale grouping (MSG) [39],

TABLE 2. Test data acquisition conditions for each posture class.

Posture class	Task	Duration	Captures
Mouse operation	Write alphabet A–Z using mouse	30 s	1
Pad operation	Write alphabet A–Z using trackpad	30 s	1
Sitting still	Sit still without moving or touching PC	10 s	1
Typing	Type alphabet A–Z on keyboard	30 s	1

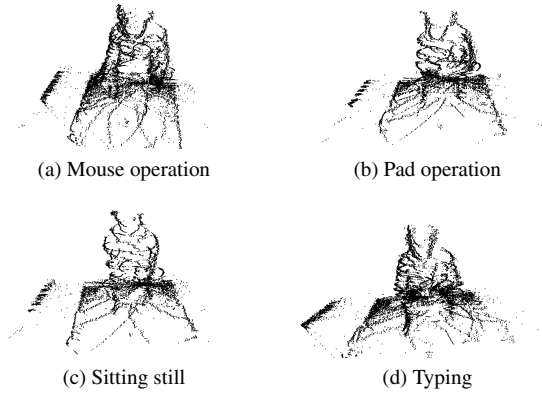


FIGURE 6. Visualization of data showing each label.

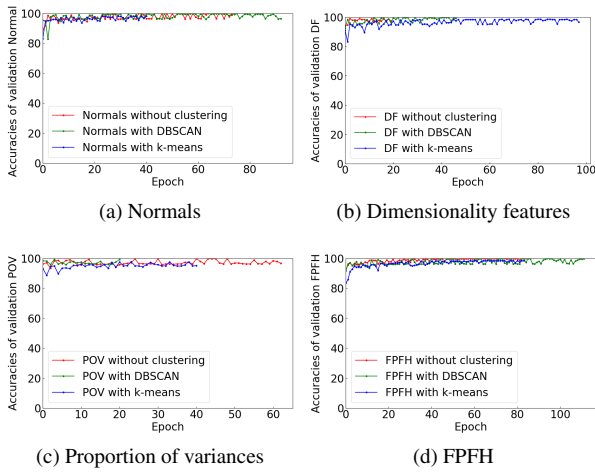


FIGURE 7. Classification accuracies of validation.

[41]. Trained classification parameters were saved after training.

E. EVALUATION METRIC

Models trained on augmented data from three subjects were evaluated on test data from twenty-eight subjects. Accuracy is defined as

$$\frac{TM + TP + TS + TT}{TM + TP + TS + TT + FM + FP + FS + FT}, \quad (1)$$

where TM , TP , TS , and TT count correct classifications for mouse operation, pad operation, sitting still, and typing, and FM , FP , FS , and FT count misclassifications.

V. EVALUATION

A. VALIDATION RESULTS

Fig. 7 plots validation accuracy per epoch for each feature with no clustering, DBSCAN, and k-means. All four features achieved nearly 100% validation accuracy across clustering methods.

Training was terminated after 15 epochs without validation-accuracy improvement beyond the best value.

B. ACCURACY

Table 3 reports test accuracy for subjects A–AB across twelve feature–clustering combinations. All cases exceeded the 25% baseline for four posture classes. No clustering or DBSCAN tended to outperform k-means; normals achieved higher accuracy without clustering or DBSCAN.

VI. CONCLUSION

An edge computing system was proposed that classifies postures with hand movement in offices using a LiDAR sensor network and point-cloud shape features.

Experiments on 28 test subjects confirmed accuracy above the 25% baseline for four posture classes across feature and clustering combinations. No clustering or DBSCAN tended to outperform k-means; normals yielded stable accuracy and dimensionality features achieved high accuracy for unseen subjects. Validation accuracy was generally near 100% for all features; test accuracy varied with feature and clustering choice. Confusion matrices showed occasional typing–pad-operation confusion due to similar hand placements, amplified when k-means removed informative points.

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TABLE 3. Classification accuracy for each subject and feature and clustering combination.

Subject	Normals			Dim. features			Prop. variance			FPFH		
	None	DBSCAN	k-means	None	DBSCAN	k-means	None	DBSCAN	k-means	None	DBSCAN	k-means
A	0.65	0.71	0.60	0.43	0.86	0.79	0.70	0.82	0.70	0.73	0.85	0.75
B	0.98	0.99	0.88	0.94	0.97	0.92	0.98	0.91	0.82	0.99	0.96	0.89
C	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70
D	0.61	0.72	0.41	0.59	0.67	0.63	0.47	0.44	0.63	0.50	0.60	0.55
E	0.63	0.69	0.46	0.73	0.60	0.59	0.60	0.58	0.56	0.66	0.53	0.57
F	0.72	0.75	0.54	0.83	0.58	0.56	0.61	0.55	0.62	0.76	0.58	0.50
G	0.68	0.75	0.40	0.72	0.53	0.54	0.56	0.55	0.54	0.75	0.55	0.40
H	0.71	0.74	0.68	0.75	0.68	0.66	0.60	0.62	0.61	0.72	0.63	0.69
I	0.76	0.90	0.63	0.92	0.66	0.65	0.61	0.61	0.60	0.75	0.64	0.62
J	0.72	0.89	0.88	0.68	0.90	0.75	0.73	0.80	0.84	0.95	0.81	0.92
K	0.74	0.87	0.87	0.97	0.86	0.77	0.65	0.82	0.79	0.83	0.80	0.80
L	0.73	0.83	0.90	1.00	0.86	0.81	0.63	0.83	0.80	0.80	0.81	0.85
M	0.59	0.59	0.45	0.56	0.60	0.44	0.56	0.46	0.40	0.49	0.60	0.45
N	0.67	0.60	0.62	0.63	0.76	0.75	0.84	0.60	0.68	0.75	0.84	0.68
O	0.85	0.71	0.71	0.70	0.76	0.73	0.73	0.61	0.68	0.80	0.73	0.72
P	0.79	0.72	0.70	0.70	0.82	0.77	0.68	0.56	0.75	0.81	0.80	0.76
Q	0.70	0.67	0.68	0.70	0.70	0.64	0.66	0.60	0.64	0.71	0.70	0.68
R	0.90	0.88	0.92	0.72	0.99	0.84	0.84	0.89	0.78	0.80	1.00	0.87
S	0.68	0.72	0.78	0.92	0.80	0.73	0.60	0.79	0.65	0.59	0.71	0.83
T	0.93	0.87	0.70	0.70	0.79	0.67	0.79	0.70	0.71	0.85	0.91	0.74
U	0.69	0.81	0.85	0.71	0.91	0.76	0.80	0.89	0.82	0.87	0.80	0.86
V	0.62	0.62	0.82	0.74	0.64	0.66	0.61	0.57	0.70	0.54	0.57	0.66
W	0.62	0.63	0.65	0.32	0.58	0.56	0.56	0.57	0.50	0.41	0.62	0.41
X	0.61	0.65	0.47	0.36	0.64	0.48	0.53	0.60	0.47	0.41	0.57	0.43
Y	0.60	0.59	0.64	0.61	0.67	0.62	0.60	0.63	0.60	0.60	0.65	0.61
Z	0.62	0.59	0.68	0.77	0.68	0.58	0.60	0.64	0.55	0.66	0.67	0.66
AA	0.64	0.87	0.94	0.90	0.79	0.88	0.62	0.76	0.71	0.58	0.54	0.63
AB	0.70	0.90	0.96	0.82	0.94	0.87	0.68	0.94	0.83	0.87	0.72	0.86

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